Guidelines

Non–Tunnelled Central Venous Catheter and Peripherally Inserted Central Catheter

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Group - wide	Medical and Nursing
Out of Scope: This guideline does not cover the insertion of non-tunnelled catheters for Haemodialysis.		

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Introduction

This multidisciplinary guideline reflects the current best available evidence in Non–Tunnelled Central Venous Catheter and Peripherally Inserted Central Catheter insertion and management; establishing the minimum standard of practice at Sir Charles Gairdner Osborne Park Hospital Care Group (SCGOHPCG) to reduce risks and avoid complications. Relevant sections should be applied in accordance with the needs of the individual patient.

Risk Statement

Non-compliance with this guideline may:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☐	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Staff Safety	☒
Other:	☐		

Definitions

Central Venous Catheter (CVC)	A CVC is a venous catheter that is inserted via the internal or external jugular, subclavian, axillary or femoral veins and advanced until the tip is located in the distal superior vena cava (SVC). With the femoral approach, the catheter tip is advanced until the tip is located in the inferior vena cava (IVC). NOTE: In this document the term CVC refers to non-tunnelled CVCs only.
Health Practitioner	A person registered under the *Health Practitioner Regulation National Law (WA) 2010* in the health professions as a Medical Officer or Intravenous Access Nurse.
Peripherally Inserted Central Catheter (PICC)	A PICC is a venous catheter that is inserted via the brachial, basilic or cephalic veins in the upper arm and advanced until the tip is located in the superior vena cava.
Personal Protective Equipment (PPE)	Includes a variety of barriers such as gloves, apron and facial protection that can be used alone or in combination to protect mucous membranes, airways, skin and clothing from contact with infectious agents.
Proceduralist	A health care professional performing a CVC or PICC insertion who must have undergone specific training and education to perform this procedure or requires supervision during the procedure.
Supervisor	An experienced Proceduralist with a high level of competence in CVC or PICC insertion and possesses a comprehensive understanding of the management of potential complications.

Legislative Requirements

All personnel who insert and/or manage CVCs or PICCs or who prescribe or administer medications using these devices must adhere to the following:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- Pharmacy Act 2010
- Work-Health-and-Safety-Act 2020

General Safety Information

- All vascular access devices carry inherent risk to both the patient and care provider. Medical and nursing staff must consider indications for CVC and PICC devices and associated complications.[1]

- Comply with *NMHS Consent to Treatment Policy* (located on SCGOPHCG HealthPoint Policies Hub). A consent form is required prior to the insertion of a CVC or PICC except in emergency situations. Consent for anaesthesia covers insertion of all vascular access devices necessary for the patient.
- Staff obtaining consent should adhere to the principles outlined in the consent script to ensure the correct process is followed, see appendix 4
- Adhere to the five moments for hand hygiene, and the principles of aseptic technique, using the risk assessment outlined in the Aseptic Technique Policy[2] (located on SCGOPHCG HealthPoint Policies Hub).
- For each insertion, **SCGH Intravascular Device Insertion record Central Vascular Access Device (MR864.2)** or **OPH Central Vascular Access Device Insertion record (MR122)** must be commenced and placed in the medical record and be maintained for the duration of the device being in situ.
- Staff should only access a CVC or PICC if it is within their scope of practice.
- All administration lines must be clearly labelled with the route of administration, the date and time that the line is commenced. All administration lines are single use only.
- Total Parenteral nutrition (TPN) should be delivered via a dedicated lumen.
- Review and document the continued need for CVC or PICC as part of the daily medical ward round. Removal should be considered if device is no longer clinically indicated.[3]
- Avoid measuring blood pressure on the arm with PICC in situ to reduce vein trauma. If unavoidable, keep measurements to a minimum and use only manual cuff.[4,5]

SCGH Intravenous Access Service

- **SCGH Intravenous (IV) Access Service** can be contacted regarding the care of a CVC or PICC: Tel: 6457 7917 (Ext. 77917)

Procedure

1 Pre-Insertion Considerations

Medical and nursing staff must consider indications for a CVC or PICC device and associated complications.[1]

1.1 Indications for insertion[6]

- Vesicant infusates with pH <5 or >9 or osmolarity greater than 600 mOsm/L. Consult with ward pharmacist.
- Patients with vascular access issues or requiring long term medications (e.g. chemotherapy or provision of in-home IV medications).
- Intravenous (IV) therapy required for at least 7 days.
- Multiple IV requirements.
- Limited sites suitable for peripheral vascular access, including:
  - Single upper limb available for access (e.g. due to fracture, axillary clearance, CVA).
  - Obesity
- TPN
- Frequent blood sampling.

1.2 Contraindications for insertion

1.2.1 CVC:

- Severe bleeding diathesis
- Ideally, the platelet count should be more than 50×10^9/l and the INR less than 1.5 before catheter CVC insertion.[6] However, clinician's discretion is required for each individual case.

1.2.2 PICC:
- Clearance of sentinel nodes – patients with lymphoedema
- Stroke affected arm
- Patients with an AV fistula
- Patients with confirmed upper limb peripheral venous thrombosis

1.3 Patients at high risk of complications

1.3.1 History of catheter-related blood stream infections: Consider referral to **Infectious Diseases (ID)** for antimicrobial locking agents.

1.3.2 History of central catheter-related venous thrombosis: Consider referral to **Anticoagulation Nurse Practitioner** and prophylactic anticoagulation.

1.4 Other considerations

1.4.1 For patients with chronic kidney disease, refer to the Clinical Practice Guideline *Vein Selection and Preservation in Patients with Chronic Kidney Disease* located on SCGOPHCG Policies Hub and seek approval to insert CVC or PICC devices from the consulting renal physician.

1.4.2 If pre-assessment tool has been completed by the IV Access Nurse and documented within the eReferral, Health Practitioner should refer to the Vein Assessment Tool prior to insertion, see Appendix 5.

1.4.3 CVC and PICC insertion must be performed in a designated area by authorized personnel.[3,6,7]

OPH: Patients at OPH requiring CVC will be referred to an Anaesthetist. Patients at OPH requiring PICC will be referred to the IV Access Service

SCGH: (Refer to Appendix 10.1: Vascular Pathways for Insertion of CVC / PICC)
- Areas designated for CVC or PICC insertion include:
  - Emergency Department
  - Interventional Radiology
  - Intensive Care Unit (ICU)
  - Theatre / Post Anaesthetic Care Unit (PACU)
  - Ground Floor Radiology – IV Access Service
- Personnel authorised to insert a CVC or PICC include:[6,7]
  - Medical practitioner trained and assessed as competent in the procedure
  - Nurse trained and assessed as competent in the procedure
  - Nurse or medical practitioner under the direct supervision of senior, experienced medical or nursing personnel trained and assessed as competent.

1.4.4 Patients having insertion of a CVC or PICC in Interventional Radiology (**SCGH**) must have:
- A Procedural checklist
- A nurse escort to and from the department
- All relevant clinical documentation

1.4.5 Obtain valid consent prior to insertion of a CVC or PICC (see **General Safety Information** on page 2)

2 Insertion - CVC

2.1 Safety Information

2.1.1 Refer to **General Safety Information on** Page 2. Principles of surgical aseptic technique must be adhered to when inserting a CVC.[3,7,8]

2.1.2 Personnel who insert CVCs must document procedure appropriately and commence **SCGH Intravascular Device Insertion record Central Vascular Access Device (MR864.2)**, or **OPH Central Vascular Access Device Insertion record (MR122)**. Documentation needs to include:
- Device type

- Device reference number
- Internal catheter length
- External catheter length
- Catheter tip position
- Confirmation that device is safe to use
- Proceduralist name

2.1.3 Optimal catheter tip position is the lower third of the Superior Vena Cava (SVC) near its junction with the right atrium.[4,6,9]

2.1.4 Catheter tip location must be determined radiographically or by other approved technologies prior to initiation of infusion therapy.[4] Obtain chest X-ray to confirm correct catheter tip placement and exclude complications, except when CVC is inserted under fluoroscopy (X-ray guidance) or using electrocardiograph (ECG) technology.[6-8]

2.1.5 At **SCGH ICU**, in non-resuscitation situations:
- All CVC and femoral lines inserted or inserted in external areas (including external sites) must be pressure transduced to confirm intravenous placement.
- Jugular and subclavian lines require a post insertion chest X-ray to assess for insertion complications.

2.1.6 **Other than in an emergency**, the Proceduralist, supervisor or assistant should stop the procedure if asepsis is breached.

2.1.7 CVC must be secured to the skin to avoid kinking, dislodgement or fracture (refer to 2.6.3).

2.1.8 Using a sutureless securement device is preferred to suturing the device to the skin as the latter further disrupts the skin around the catheter site and can lead to inflammation. A sutureless device eliminates the risk of sharps injury to health care personnel.

2.1.9 If a patient is being referred to ICU (**SCGH**), suturing is recommended to reduce the risk of inadvertent dislodgement of CVC.

2.2 Position and Preparation of Patient

2.2.1 Proceduralist explains the procedure to the patient and obtains consent.

2.2.2 Prior to scrub, position the patient lying flat and straight in the bed as close to the edge / head of bed as possible, clear wires and tubes to allow access to the optimal position.

2.2.3 Clear the area of nasogastric tubes, oxygen mask ties, endotracheal tube ties, ECG leads.

2.2.4 Tilt bed head down by 15-30 degrees, this is very important for 2 reasons:
- minimises the risk of air embolism
- makes the vein firmly distended and easier to puncture, without going right through it.

Positioning for INTERNAL JUGULAR lines:
- Head straight not flexed at the neck
- Tilted head 45 degrees to the opposite side

Positioning for SUBCLAVIAN lines:
- Arms at the side, pulled down so clavicle roughly 90 degrees to sternum
- Consider rolled towel between shoulder blades

2.3 Equipment Requirements

- Ultrasound machine, sterile transducer gel, and sterile probe cover
- Sterile dressing pack including full body sterile drape, sterile gown and sterile labels[2,3,6]
- Full PPE (including surgical mask, theatre hat, and facial protection) [2,3,6]
- Sterile gloves[3]
- CVC insertion kit with appropriate number of lumens (Refer to patient treatment plan)
- 10ml luer lock syringes (one per lumen)
- 10ml sodium chloride 0.9% for injection (prime each lumen before insertion)[10]
- 1 x 26G needle
- 5ml syringe

- Lignocaine 1%
- Chlorhexidine 2% in Alcohol 70% solution unless contraindicated by allergy / sensitivity[2-4,10,11] (Alternatively, Povidone-iodine 5% in Alcohol 70%, or Prontosan Solution, can be used if allergies present)
- Sterile gauze
- Suture kit & suture material
- Tegaderm CHG™, a BIOPATCH® antimicrobial chlorhexidine impregnated sponge and sterile, occlusive, and transparent dressing, or a Kendall antimicrobial impregnated sponge and sterile, occlusive, and transparent dressing. (e.g. OPSITE IV3000®)[2]
- Positive Pressure Valves (PPV) (e.g. MaxPlus®) - one per lumen[12]
- Absorbent sheet
- Clinical waste bin
- **SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR 864.2)** or **OPH Central Vascular Access Device Insertion record (MR122)**

2.4 Procedural Information

2.4.1 Don hat and surgical mask. Perform a surgical scrub using Chlorhexidine 4% antiseptic hand wash solution

2.4.2 Dry hands using disposable linen towel supplied in the CVC pack. Don sterile gown and sterile gloves from the CVC trolley

2.4.3 Proceduralist to initiate team time out (TTO) to ensure the activities of each team member are coordinated and to verbally verify the following details:

- The presence of the correct patient.
- The correct type of procedure to be performed.
- The correct procedure side and site has been identified / marked with water-resistant marker or documented on consent and X-ray request.
- Prior to insertion, the skin should be prepared with Chlorhexidine 2% in Alcohol 70% thoroughly and allow to air dry (unless contraindicated by allergy / sensitivity[2-4,10,11] (Alternatively, Povidone-iodine 5% in Alcohol 70%, or Prontosan Solution, can be used if allergies present).

INTERNAL JUGULAR VEIN APPROACH

2.4.4 Stand at the head of the bed

2.4.5 Administer adequate local anaesthetic

ULTRASOUND METHOD (Recommended)

- Using ultrasound for guidance, locate a picture with the carotid medial to the internal jugular vein (IJV), the most common error is to transfix the vein; and if the carotid is directly behind the vein, the needle will end up in it. This is best achieved with the probe vertical (perpendicular to the bed, anteriorly on the neck).

Always keep the tip of the needle in view as you advance, there are 3 ways to do this:

1. Continuously slide the ultrasound probe or angle it with the advancing needle tip.
2. Longitudinal “in plane” imaging, beware of inadvertently moving over the adjacent artery.
3. Walk the needle tip down the ultrasound field.

NOTE:

- The most common error is seeing the shaft of the needle and thinking it is the tip when the tip is way past the plane of ultrasound.
- Don’t advance unless you can see the tip

LANDMARK METHOD (when ultrasound guidance is not available)

- Locate mid-point of anterior edge of sternomastoid, feel sternal head and mastoid process and start at the mid-point, or lateral to the cricoid cartilage.
- Angle needle 45 degrees to the skin.
- Once skin is punctured, CONTINUOUSLY ASPIRATE while advancing the needle.
- Aim at the IPSILATERAL NIPPLE; this is critical as it is aimed away from the carotid and central neck structures.

2.4.6 Once the vessel is cannulated, visualise the guidewire in the vein before dilating, i.e. follow the wire down to be sure it has not passed through the vein

SUBCLAVIAN VEIN APPROACH

ULTRASOUND METHOD (Recommended)

- The axillary vein will be seen using the ultrasound probe (not the subclavian), identify the vein caudal to the artery.
- Give enough room to be able to follow the tip of the needle into the vein.
- Depending on the depth, a fairly steep angle to the skin will be needed to hit the vein.
- Remember the vessels lie on the rib cage, because of the steep angle and the proximity of the vein to the chest wall - **DO NOT** advance the needle unless you can confidently see the tip of the needle, to avoid the risk of causing a pneumothorax.

LANDMARK METHOD (when ultrasound guidance is not available)

- Identify the junction of the medial third and the middle third of the clavicle, start there and aim at the sternal notch or 1 cm above it, hugging the underside of the clavicle, always aspirate as you are advancing.

2.4.7 Cannulate the vein. Use the cannula (preferable) or the introducer needle attached with a syringe, when cannulating, insert guidewire and remove the cannulating needle (NB: Potential for arrhythmias when using internal jugular and subclavian veins). The floppy end of the guidewire must always enter the vessel first. It is important to constantly hold on to the guidewire.

2.4.8 Confirm the position of the guidewire by ultrasound scanning the vein (except subclavian venous access). This enables the Proceduralist to visualise the guidewire inside the vein from the insertion point until the most distal part of the vein that can be scanned. This is to be done prior to dilating and inserting the catheter in the blood vessel. If still in doubt, seek senior medical opinion.

2.4.9 Using a scalpel to make an incision over the skin at the entry site.

2.4.10 Slide the dilator over the fixed guidewire to avoid kinking using a twisting motion. Advancing both together can result in kinking and creating a separate track outside the vessel. (Dilators are to dilate the soft tissues and not the vessels, so do not use the entire length of the dilator).

2.4.11 Remove the dilator whilst the guidewire remains in situ

2.4.12 Advance CVC over guidewire, making sure to secure the distal end of the guidewire.

2.4.13 Once the CVC is advanced to the required level, remove the guidewire.

2.4.14 Check ability to aspirate blood from all lumens.

2.4.15 Attach a PPV for each lumen.

2.4.16 Lock the lumens with sodium chloride 0.9% using a push pause technique.

2.4.17 Clean catheter and insertion site of blood with skin antiseptic, and allow to air dry, prior to applying dressing and securement device.

2.4.18 Apply Chlorhexidine-impregnated dressing, CHG gel semi-permeable dressing (Tegaderm CHG™) or Chlorhexidine-impregnated sponge dressing (e.g. BIOPATCH®) to insertion site (correct side up, ensuring edges meet, split lined up with the catheter and has complete contact with the skin).

2.4.19 Apply gauze if any bleeding or oozing.

2.4.20 Attach a transducer to the distal lumen of the CVC line for further confirmation of intra venous placement of the CVC line.
2.4.21 If CVC is inserted under fluoroscopy and catheter tip placement has been confirmed, connect lines as indicated.
2.4.22 Care must be taken not to contaminate the lines when connecting infusions, the transducer or dialysis / hemofiltration lines (e.g. tip of the transducer onto the sterile field and hand the end of the transducer to the assistant to connect to the flush bag as appropriate).
2.4.23 Remove PPE.
2.4.24 Discard rubbish, and safely dispose sharps.
2.4.25 Perform hand hygiene.
2.4.26 Complete documentation on the **SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR 864.2)** or **OPH Central Vascular Access Device Insertion record (MR122).**

Blood gas sampling is **NOT** recommended as a method of differentiation between venous or arterial blood.

The tip of the CVC should be in the SVC for neck lines and in the inferior vena cava for femoral lines. As a general guide, insert the catheters to desirable length: 12cm to 18cm marking for right sided neck lines; and 20cm marking for the left sided neck lines and femoral lines. The position of the tip of the CVC line must be confirmed by X-ray or real-time fluoroscopy (except for femoral venous access).

2.5 Post Insertion

2.5.1 Obtain a chest X-ray to exclude complications associated with insertion of the CVC.
2.5.2 A chest X-ray is not required if the device has been inserted via Interventional Radiology.
2.5.3 Catheter tip location must be determined radiographically or by approved ECG technology (e.g. Nautilus Delta or Sherlock 3CG) prior to initiation of infusion therapy.
2.5.4 Catheter tip location must be documented in the patient's medical record and the **SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR 864.2)**, or **OPH Central Vascular Access Device Insertion record (MR122).**
2.5.5 Educate the patient regarding the device, its care and maintenance when appropriate.

2.6 Avoiding Complications during CVC Insertion

2.6.1 AVOIDING CATHETER ASSOCIATED INFECTION

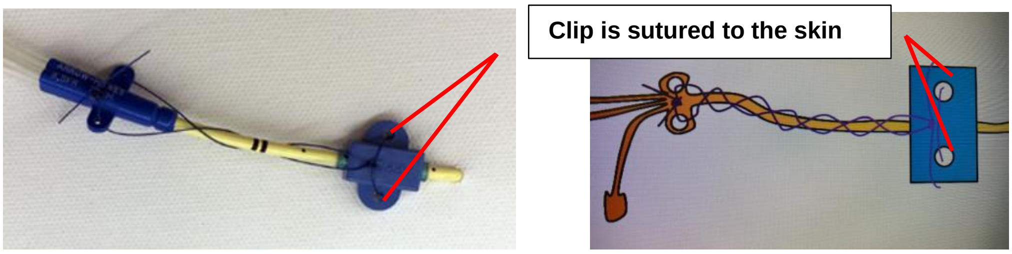
- If clinically appropriate use supra-diaphragmatic site of CVC insertion.
- Perform a surgical scrub using Chlorhexidine 4% antiseptic hand wash solution.
- Maximum sterile barrier precautions (surgical mask, theatre cap, sterile gloves, gowns and large drapes) and aseptic techniques have been shown to lower the risk of acquiring catheter associated infections.[6]
- Prior to insertion, the skin should be prepared with Chlorhexidine 2% in Alcohol 70% thoroughly and allow to air dry.
- Secure with Prolene (polypropylene) / Ethilon (nylon) sutures.
- Apply transparent Chlorhexidine-impregnated dressings following sutures to secure the catheter.

2.6.2 LOST WIRE

- Seek senior clinical assistance, document and urgently contact Interventional Radiology and / or vascular team for advice.

2.6.3 AVOIDING CVC DISLODGEMENT

- Suture / clip correctly using a 2-point fixation system.
- Use at least size 0 nylon / polypropylene sutures, deep 1 cm bite of skin.
- Suture the clip to the skin, and the CVC itself is anchored to the clip using the sandal strapping technique illustrated below:



- Apply transparent Chlorhexidine-impregnated dressing over the insertion site following suturing.
- Consider Fixamol® perimeter to the dressing.

2.6.4 AVOIDING DAMAGE TO SURROUNDING STRUCTURES

- Consider the vein, its size, depth and trajectory – decide on the best approach.
- Consider surrounding structures – superficial, adjacent and deep to the target.
- Plan the approach to mimic the landmark technique – going out the back of the IJV to avoid hitting the important structures lying behind the vein.
- Carefully and slowly follow the tip of the needle with ultrasound as it approaches the vein.
- Watch the front of the vein indent, advance and then confirm the needle tip in the vein.

2.6.5 AVOIDING ARTERIAL CANNULATION

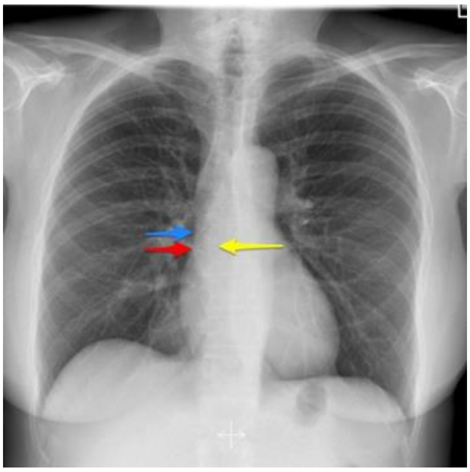
- Identify vein and artery using ultrasound – ensure the artery is not in the needle trajectory (including deep to the vein); recall the subclavian artery passes deep to the low IJV.
- Observe the nature of blood flow upon needle insertion.
- Use ultrasound to confirm intravenous position of guidewire – follow distally to ensure the wire does not go through the back of the vein.
- If in doubt, do not dilate, insert a cannula over the guidewire and attach pressure transducer to differentiate arterial from venous pressures.
- In the event of needle puncture of an artery, remove the needle and apply direct pressure for 15 minute.
- Inadvertent arterial cannulation – leave device in situ. **DO NOT** remove. **DO NOT** use the line. Urgently contact Interventional Radiology and / or vascular team for advice.

2.6.6 AVOID ARRHYTHMIA

- Ensure patient on cardiac monitor and monitor visible during procedure.
- In the event of arrhythmia, withdraw guidewire a few centimetres, reassess patient.

2.6.7 AVOID INCORRECT INTRAVENOUS PLACEMENT

- The tip of a CVC placed in the IJV or subclavian vein should sit in the lower 1/3 of the SVC or at the cavo-atrial junction.
- Normal chest X-ray demonstrating different methods of estimating the position of the cavo-atrial junction:
  - Yellow arrow: two vertebral body levels below the level of the carina
  - Blue arrow: intersection of bronchus intermedius with superior right heart border
  - Red arrow: intersection of the superior right heart border with SVC contour



Note: If the line is elsewhere it will need repositioning – consider asking Interventional Radiology to reposition a difficult catheter.

2.6.8 AVOID OTHER COMPLICATIONS

- Never let go of the wire – ensure it is always seen either proximal or distal to the CVC while inserting the CVC.
- Check the chest X-ray for pneumothorax post placement.
- Where possible avoid CVC in coagulopathic patients – the femoral site is more compressible if CVC placement is essential in these patients, consider a PICC.

3 Insertion – PICC

3.1 Safety Information

Refer to **General Safety Information on** Page 2. Adhere to surgical aseptic technique when inserting a PICC.[3,7,8] Appropriate PPE must be used to prevent contamination and mucosal or conjunctival splash injuries.

3.2 Equipment Requirements

- Ultrasound machine, sterile transducer gel, and sterile probe cover
- Sterile dressing pack including full body sterile drape, sterile gown and sterile labels[2,3,6]
- Full PPE (including surgical mask, theatre hat, and facial protection)[2,3,6]
- Sterile gloves[3]
- 1 x 26G needle
- 20G cannula
- PICC insertion kit with appropriate number of lumens (Refer to patient treatment plan)
- 5ml syringe
- Lignocaine 1%
- 10ml luer lock syringes (one per lumen)
- 2 x 10ml sodium chloride 0.9% for injection (prime each lumen before inserting)[10]
- Chlorhexidine 2% in Alcohol 70% solution or Prontosan solution.[2-4,10,11]
- Sterile gauze
- Tegaderm CHG, a BIOPATCH® antimicrobial chlorhexidine impregnated sponge and a sterile, occlusive, and transparent dressing or a Kendall antimicrobial impregnated sponge and a sterile, occlusive, and transparent dressing. (e.g. OPSITE IV3000®) [2]
- Statlock®
- Positive pressure valves (PPV) (e.g. MaxPlus®) - one per lumen[12]
- Clinical waste bin
- Absorbent sheet
- **SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR 864.2)** or **OPH Central Vascular Access Device Insertion record (MR122)**

3.3 Positioning and Preparation of Patient

3.3.1 Inform patient of procedure as appropriate

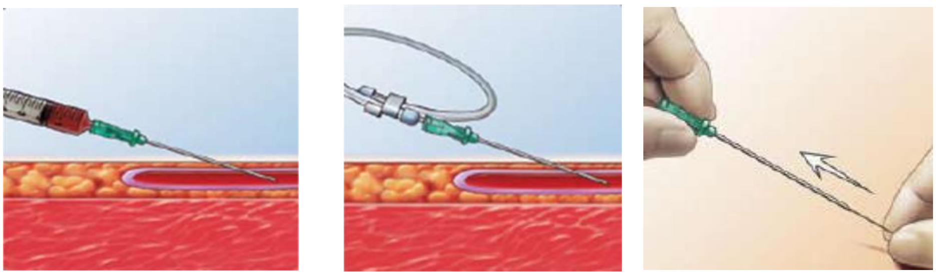
3.3.2 Check for known allergies
3.3.3 Check consent form and X-ray request form, or online X-ray request iscompleted (including specific site as appropriate)
3.3.4 Perform hand hygiene
3.3.5 Position patient supine with selected arm exposed and abducted at 90 degrees.[7] Utilise arm board if needed.
3.3.6 Clip hair around selected insertion site if excessive.
3.3.7 Place waterproof protective sheet (Bluey®) under insertion site.
3.3.8 Apply transducer gel to ultrasound probe.

3.3.9 Using the ultrasound to choose optimum insertion site, and perform vein assessment.
3.3.10 Once site is chosen, clean off transducer gel and ensure patient is comfortable.
3.3.11 Perform hand hygiene

3.4 Insertion Procedural Information

3.4.1 Don hat and surgical mask. Perform a surgical scrub using Chlorhexidine 4% antiseptic hand wash solution.
3.4.2 Dry hands using disposable linen towel supplied in the sterile dressing pack. Don sterile gown and sterile gloves from PICC trolley.
3.4.3 Proceduralist to initiate team time out (TTO) to ensure the activities of each team member are coordinated and to verbally verify the following details:
- The presence of the correct patient.
- The correct type of procedure to be performed.
- The correct procedure side and site has been identified / marked with water-resistant marker or documented on consent and X-ray request.

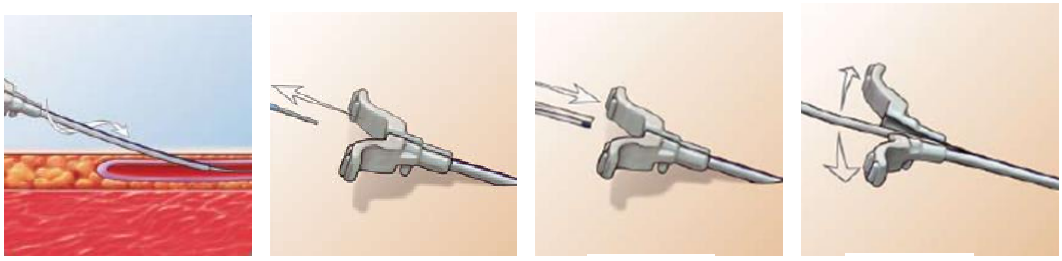
3.4.4 Ask Assistant to apply tourniquet to upper arm and tighten.
3.4.5 Clean the patient’s arm with Chlorhexidine 2% in Alcohol 70% solution or Prontosan solution using a circular motion beginning at the centre and moving to the periphery, covering the patient’s entire upper arm. Allow area to air dry completely.
3.4.6 Remove the white tape from the back of the drape and apply the sterile drape to the patients arm, instructing them not to touch the sterile field. Apply the sterile probe cover to the ultrasound probe with the help of the assistant.
3.4.7 Draw up Lignocaine 1% 50mg in 5ml syringe. Attach a 25G subcutaneous needle and administer local anaesthetic superficially around the intended PICC insertion site.
3.4.8 Cannulate the patient using real time ultrasound with a 20G peripheral IV cannula and release the tourniquet once IV access (venous return) is obtained.



3.4.8 **3.4.9** **3.4.12**

3.4.9 Insert the 45cm guidewire into the patient’s vessel via the IV cannula. Ensure to leave approximately 15cm of wire external to avoid accidental loss of the wire into the patient.
3.4.10 Prime the PICC line with sodium chloride 0.9% and insert the 80cm guidewire into the PICC with the soft end of the wire towards the tip of the line. Pull the wire back so that no wire is protruding from the end of the line, and clamp it in place.
3.4.11 When inserting a double lumen PICC line, prime the proximal lumen with sodium chloride 0.9% and attach a PPV (e.g. Posiflow®). Prime the distal lumen and insert the 80cm wire as instructed for the single lumen line. ***Note**: The 80cm wire will not fit down the proximal lumen of a double lumen PICC line.

3.4.12 Hold the 45cm guidewire in place and remove the IV cannula from the patient's arm.
3.4.13 Insert the 10cm tissue dilator with peelable sheath over the guide wire in the patient's arm. Ensure that there is approximately 10cm of wire visible at all times during this step. Using a pushing or twisting motion, insert the tissue dilator with peelable sheath through the subcutaneous tissue into the patient's vessel.



3.4.13 3.4.16 3.4.17 3.4.20

3.4.14 Remove the 45cm guidewire from the centre of the tissue dilator.
3.4.15 Instruct the patient to turn their head to the side of insertion and tuck their chin into the clavicle to help minimise chance of internal jugular placement.
3.4.16 Remove the tissue dilator, leaving the peelable sheath in the vein.
3.4.17 Thread the catheter slowly through the peelable sheath. If you meet resistance, retract the catheter, reposition arm, and reattempt to insert the catheter. Lay the patient completely flat to allow for smooth insertion.
3.4.18 Remove the wire from inside the PICC line and attach a 10ml syringe of sodium chloride 0.9%. Check the PICC is not positioned in the patient's IJV by using ultrasound to assess the vessel as you flush the catheter. Ask the patient to listen for any sounds or sensations of water in their neck or ear whilst flushing. Attach a primed PPV to the end of the lumen. The PPV should be primed prior to being attached to the device with 0.9% Normal Saline.
3.4.19 Use the measuring tape and landmarks, estimate the length of PICC line to remain internal. Position the catheter at the estimated length and test the PICC is aspirating and flushing well.
3.4.20 Pull the tabs on the sheath apart and remove from patient's vessel whilst holding the PICC line in place.
3.4.21 Apply the extra wings from the PICC pack, if necessary, use Chlorhexidine 2% in Alcohol 70% solution or Prontosan solution to prepare the skin, then allow the skin to dry, and secure the line into the Statlock®. Attach the Statlock® to the patient's skin, leaving approximately 2cm between the insertion site and securement.
3.4.22 Dress the PICC line with either a Tegaderm CHG™, a BIOPATCH® antimicrobial chlorhexidine impregnated sponge and a sterile, occlusive, and transparent dressing, or a Kendall antimicrobial impregnated sponge and a sterile, occlusive, and transparent dressing, ensuring that any exposed catheter is secured under the dressing. Apply a pressure dressing to the site if there is any bleeding around the insertion site.
3.4.23 Confirm that both guidewires have been removed and are intact.
3.4.24 Dispose of sharps and waste.
3.4.25 Remove PPE.
3.4.26 Perform hand hygiene.
3.4.27 Document on the e-referral information regarding the PICC line insertion. Document any difficulties / complications during insertion.

3.5 Post Insertion

3.5.1 Send the patient for a chest X-ray to confirm tip position. Document confirmation that PICC is safe for use - this may be recorded on iSOFT Clinical Manager, e-referral or a procedure report, and **SCGH Intravascular Device Insertion record Central Vascular Access Device (MR864.2)** or **OPH Central Vascular Access Device Insertion record (MR122)**.
3.5.2 **SCGH ICU**: Transduction of PICC to confirm placement is at the discretion of the SCGH ICU Medical officer.

3.5.3 Educate the patient on how to care for the PICC line. If they are an outpatient, educate them on who they can contact if they notice any redness, tenderness, swelling or other complications with their line. Provide the patient with information brochures.

4 Insertion Site Dressing

4.1 Safety Information

4.1.1 Refer to General Safety Information on page 3.
4.1.2 Loop the CVC or PICC tubing distally and secure to avoid catheter kinking, dislodgement or fracture.

4.2 Management

Assess insertion site and associated limb at least once per shift, prior to any access, at dressing changes or as clinically indicated.[3-5]
Assess insertion site and associated limb for:

- Tenderness, heat, pain, erythema
- Discharge
- Swelling
- Suture / or securement device and dressing integrity
- Position, to exclude any migration
- Patency

4.2.1 Change transparent dressings and PPV every 7 days or sooner if no longer intact or if moisture collects under the dressing. [2-5]
4.2.2 Clean insertion site with Chlorhexidine 2% in Alcohol 70% solution unless contraindicated. For patients with Chlorhexidine allergy, consider Povidone-iodine, or Prontosan Solution, or sodium chloride 0.9%.[2-5,8,10,11]
4.2.3 Apply either a Tegaderm CHG™, a BIOPATCH® antimicrobial chlorhexidine impregnated sponge and sterile, occlusive, and transparent dressing or a Kendall antimicrobial impregnated sponge and a sterile, occlusive, and transparent dressing .[2]
4.2.4 Use gauze dressing when patient is:

- Diaphoretic[13]
- Allergic to occlusive type dressings
- Change gauze dressing 24 hourly to visually inspect the site.[4,5]

4.2.5 In case of excessive bleeding from insertion site:

- Apply pressure dressing with gauze and tape. Remove gauze pressure dressing after 24 hours and replace with a Tegaderm CHG™, a BIOPATCH® antimicrobial chlorhexidine impregnated sponge and a sterile, occlusive and transparent dressing or a Kendall antimicrobial impregnated sponge and a sterile, occlusive, and transparent dressing (e.g. OPSITE IV3000®)[2,3,155]

4.2.6 Report soiled or damaged sutures or securement device to [SCGH IV Access Service]
4.2.7 Cover insertion site with waterproof plastic covering during showering to prevent the dressing becoming loose and wet.

Note: PPV changes can occur at same time as dressing change.

4.3 Procedural Information

4.3.1 **Equipment Requirements**

- PPE including non-sterile gloves and sterile gloves
- Wet strength bag
- Dressing pack
- Chlorhexidine 2% in Alcohol 70% solution or Prontosan solution[2,10,11]
- Tegaderm CHG™ , a BIOPATCH® antimicrobial chlorhexidine impregnated sponge and a sterile, occlusive, and transparent dressing or a Kendall antimicrobial impregnated sponge and a sterile, occlusive, and transparent dressing (e.g. OPSITE IV3000®)[2,3,15]

- Securement device, if required for PICC (e.g. Statlock® / Grip-Lok®).
- Skin Barrier wipe (e.g. Cavillon No Sting Wipe) if required.
- PPV as indicated[12]

4.3.2 **Procedure**

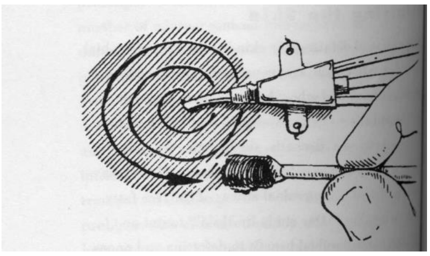


Figure 1 Correct technique to clean insertion site

- Confirm patient identity and explain the procedure.
- Perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes. Allow time to dry.
- Gather all equipment.
- Perform hand hygiene.
- Prepare the sterile field.
- Open dressing pack using the corners and add equipment to the sterile field using non touch technique.
- Put on PPE (non-sterile gloves).[2]
- Use non-touch technique to remove old dressing.
- Remove gloves.
- Perform hand hygiene; put on sterile gloves.
- Clean around catheter insertion site with Chlorhexidine 2% in Alcohol 70% or Prontosan solution using aseptic non touch technique[2,10,11,14] as shown in **Figure 1**. Allow to air dry for 30 seconds.
- Report loose PICC sutures or tender, red suture site to medical officer or SCGH IV Access Service. Consider replacing sutures with a securement device (e.g. StatLock® or Grip-Lok®).
- Measure external PICC length from insertion site to integral wings at each dressing change (See **Figure 2**). If suspected catheter dislodgment, continue dressing and notify SCGH IV Access Service

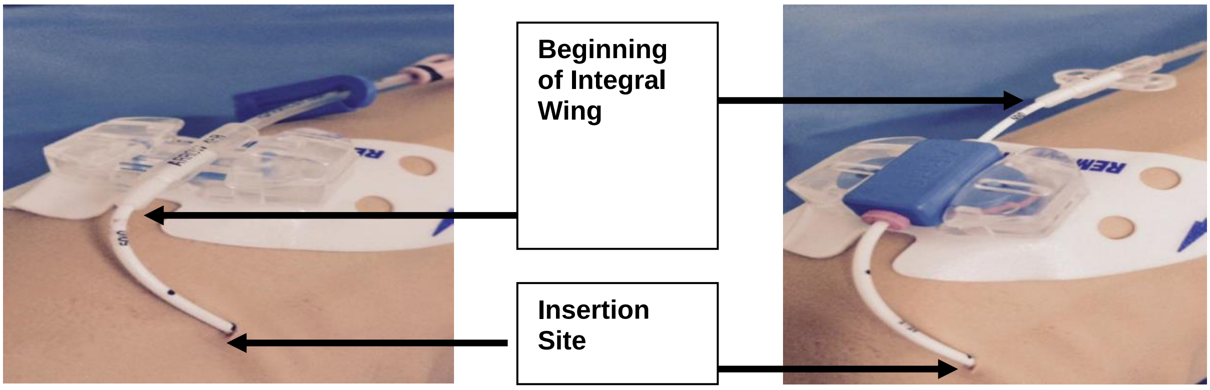


Figure 2 Measuring PICC external catheter length

- If changing the securement device: Secure it to skin, approx. 2-3cm from insertion site. Loop and secure any excess external catheter with additional securing tape (**Figure 3**).

Figure 3 Correct Statlock® application

Figure 4 Correct Tegaderm CHG™ dressing application

- Apply a Tegaderm CHG,™ (**Figure 4**) a BIOPATCH® antimicrobial chlorhexidine impregnated sponge around the catheter entry site, with blue side up and white foam side next to the patient's skin[3] then apply a sterile, occlusive, and transparent dressing (**Figure 3**) or a Kendall antimicrobial impregnated sponge and a sterile, occlusive, and transparent dressing. In addition, a Fixomul® window frame may be created around the dressing to increase stability.[2,14,15]
- Dispose of sharps and waste appropriately.
- Remove sterile gloves and perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes.
- Perform hand hygiene.
- Document on either the **SCGH Intravascular Device Insertion record Central Vascular Access Device (MR864.2)** or **OPH Intravascular Device Insertion Record (MR122)**, and medical record.
- If suspected catheter migration: Inform medical officer and SCGH IV Access Service. Chest X-ray may be required to confirm catheter tip position and device safety.

5 Changing Positive Pressure Valves (PPV)

5.1 Safety Information

5.2.1 Refer to **General Safety Information** on **Page 3**
5.2.2 Do not remove side clamps.
5.2.3 Clamp CVC or PICC prior to changing PPV. Leave clamps open at all other times to reduce catheter occlusion and infection.[4,5]
5.2.4 Wear PPE as per standard precautions.

5.2 Management

5.2.5 Assess PPV for patency and presence of visible blood at least once per shift.[3-5]
5.2.6 Change PPV every 7 days or sooner if leaking or contaminated with visible blood.[3-5]

5.3 Procedural Information

5.3.1 Equipment Requirements

- PPE including non-sterile gloves and sterile gloves
- Wet strength bag
- Dressing pack
- Chlorhexidine 2% and Alcohol 70% wipes
- PPV (one for each lumen)[12,16]
- 10ml syringes and sodium chloride 0.9% (one for each lumen)[10,17]
- Drawing up needle
- Sterile gauze

5.3.2 Procedure

- Confirm patient's identity and explain the procedure.
- Perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes.
- Gather equipment.
- Perform hand hygiene and don PPE.
- Open dressing pack using the corners. Add items required using non touch technique.
- Place sterile sheet under the CVC or PICC lumens.
- Put on non-sterile gloves.
- Close side clamps and remove old PPV using non-touch technique.
- Remove gloves.
- Perform hand hygiene and don sterile gloves.
- Hold the end of the lumen using sterile gauze

- Scrub the surface of each lumen for 30 seconds with sterile gauze soaked in Chlorhexidine 2% and Alcohol 70% wipes using friction.
- Allow to air dry for at least 30 seconds.
- Prime PPV with sodium chloride 0.9%. Do not disconnect syringe.
- Attach primed PPV with syringe intact.[12]
- Release side clamp.
- Flush each lumen with sodium chloride 0.9% using a rapid, pulsating push/pause positive pressure technique. Do not entirely empty flush volume (Disconnect the syringe before flush volume empties 'bottoming out') to reduce reflux of blood back into the lumen, decreasing risk of catheter occlusion.[4,5,10,17,18]
- Dispose of all sharps and waste appropriately.
- Remove PPE, including sterile gloves.
- Perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes.
- Perform hand hygiene.
- Document PPV changes on the **SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR864.2)** or **OPH Intravascular Device Insertion Record (MR 122)** and / or medical record.

6. Accessing and Flushing Lumens

6.1 Safety Information

6.1.1 Refer to **General Safety Information** section: Page 3

6.1.2 PPE must be used when accessing and flushing CVC and PICCs.

6.1.3 Flush CVC and PICC devices to maintain patency when line is not in continual use.

6.1.4 Use only a 10ml luer lock syringe when flushing a CVC or PICC to reduce the risk of catheter rupture.[17] Use a 10ml syringe or larger when administering medications.

6.1.5 Immediately flush CVC and PICCs at completion of infusion therapy, and upon disconnection from infusion pumps.[4,5]

6.1.6 Flush each lumen with Sodium Chloride 0.9% using a rapid, pulsating push/pause positive pressure technique. **Do not entirely empty flush volume** (Disconnect the syringe before flush volume completely empties 'bottoming out') to reduce reflux of blood into lumen and catheter occlusion.[4,5,10,17,19]

6.1.7 If there is visible blood or other solutions (e.g. TPN) in positive pressure valve (PPV) or catheter lumen, multiple flushing or valve change will be required.[4,5]

6.1.8 If strong resistance is met during flushing:

- Notify either the ward CNM/S, SDN, SCGH IV Access Service, After Hours CNM/S or Duty Anaesthetist for assistance.

6.1.9 Use PPV at all times to ensure a closed system and reduce reflux of blood into catheter, catheter occlusion, and infection.[4,5]

6.1.10 PPV are part of a needleless system. Never use needles with PPV to avoid damage to valves, increasing the risk of catheter occlusion and infection.

6.1.11 Designate one lumen for each non-compatible infusion e.g. TPN, inotropic drugs, chemotherapy.[4,5]

6.1.12 Flush CVC or PICC with 10ml Sodium Chloride 0.9% or BD Posiflush™ pre-filled Saline syringe every 24 hours or as necessary when not in regular use.[4,5,10,17,18,20]

6.2 Procedural Information

6.2.1 Equipment Requirements

- Kidney dish
- PPE including non-sterile gloves
- Wet strength bag
- 1 x large 70% Alcohol impregnated wipe
- Sodium Chloride 0.9%

- 10ml luer lock syringe and drawing up needle or BD Posiflush™ Pre-filled Saline Syringe[10,17,18]

6.2.2 **Procedure**

- Perform hand hygiene.
- Gather equipment and place in a clean kidney dish.
- Perform hand hygiene.
- Apply non-sterile gloves.
- Prepare IV substances using a non-touch technique and protecting key parts.
- Prior to every access, scrub the PPV with a large 70% Alcohol impregnated wipe for 30 seconds using different areas of the wipe. Allow to air dry for 30 seconds..[8,16]
- Aspirate a small amount of blood to assess catheter patency.
- Flush lumens before and after each access.
- Flush each lumen with Sodium Chloride 0.9% using a rapid, pulsating push/pause positive pressure technique.[20] Do not entirely empty flush volume (Disconnect the syringe before flush volume completely empties 'bottoming out'), to reduce reflux of blood back into the lumen and catheter occlusion.[4,5,10,17-19]
- When transfusing blood products, flush CVC or PICC catheter with 20mls Sodium Chloride 0.9% (2x10ml BD Posiflush™ Pre-filled Saline Syringes) following disconnection of the infusion set.[10,17,18]
- Dispose of sharps and waste appropriately.
- Remove gloves and perform hand hygiene.

Note: If resistance is met when flushing CVC or PICC with BD Posiflush™ Pre-filled Saline Syringe, try using 10ml luer lock syringe with 10mls Sodium Chloride 0.9%, to enable better positive pressure technique.

7. Blood Sampling

7.1 Medical Requirements

Pathology blood sample forms should clearly state that the blood sample is from a CVC or PICC.

7.2 Safety Information

7.2.1 Refer to **General Safety Information** section: Page 2.
7.2.2 Sampling blood from CVC or PICC increases risk of catheter occlusion.
7.2.3 Ideally blood samples should be taken from peripheral veins; however a CVC or PICC is often inserted because a patient has poor peripheral access, making blood sampling from a CVC or PICC necessary.
7.2.4 To maintain a closed system and reduce the risk of infection and contamination, DO NOT remove the positive pressure valve (PPV) during blood sampling.[3]
7.2.5 To help minimise the risk of catheter occlusion the preferred method is to aspirate samples with a syringe and transfer into appropriate vials using the pink transfer vacutainer®.
7.2.6 Vigorous pulsative saline flushes immediately post sampling is essential to reduce occlusion risk.
7.2.7 Continuous infusions can be paused for blood sampling from alternate lumens, where appropriate, but must not be disconnected to obtain blood samples. Use alternative route.
7.2.8 All specimen tubes require 3 points of identification. Label after collection.
7.2.9 Specimens must be labelled by the collector at the bedside.
7.2.10 Clearly label each specimen with patient's full name, UMRN, date of birth, time and date of collection, and collector's signature. Use patient's hospital label when possible.
7.2.11 Blood culture bottles also require information about sampling site (i.e. peripheral, CVC or PICC).
7.2.12 Specimens without patient's full name and date of birth will be rejected by laboratory.

Refer to Transfusion Policy Manual: Sample Collection and Labelling Guideline located on SCGOPHCG HealthPoint Policies Hub for more information.

7.3 Procedural Information

7.3.1 Equipment Requirements

- 2 x Kidney dish
- Non-sterile gloves
- Blood specimen tubes
- Large 70% Alcohol impregnated wipe
- 1 x 10ml luer lock syringe (for 5-10ml blood discard, **do not discard if taking blood cultures)**
- 1 x 10ml or 20ml luer lock syringe (syringe size determined by sample requirement)[10, 17]
- 20ml sodium chloride 0.9%
- 10ml luer lock syringe and drawing up needle or 2x BD PosiFlush™ 10 ml Pre-Filled Saline Syringes (for flushing post sampling)[10]
- PINK Vacutainer transfer device and luer lock adapter, or BLUE Vacutainer access device

7.3.2 PROCEDURE

- Confirm patient's identity and explain the procedure.
- Perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes.
- Gather equipment and check expiry dates.
- Turn off infusions one minute prior to sampling if possible.
- Where possible, use distal lumen (which has the largest bore) to obtain sample.[21]
- Perform hand hygiene and apply non-sterile gloves.
- Scrub the PPV using friction for 30 seconds with a large 70% Alcohol impregnated wipe. Allow to air dry for 30 seconds before accessing.[10,16]
- Attach 10mL syringe to PPV and aspirate 5-10mls of blood. Discard syringe and blood to prevent an abnormal reading from contaminated blood.[21]

NOTE: If taking blood cultures DO NOT discard first 5-10mls. Refer to SCGH Infection Prevention and Control Manual, Specimen Collection for Microbiological Examination Guideline located on SCGOPHCG HealthPoint Policies Hub for more information.

Preferred sampling method using syringe transfer to vacutainer® (as this decreases catheter occlusion risk):

a. Attach sample syringe to PPV and aspirate required blood volume.
b. Place syringe in clean kidney dish.
c. Immediately following blood sampling, scrub the PPV using friction for 30 seconds with a large 70% Alcohol impregnated wipe and flush with 20ml sodium chloride 0.9% using 2 x 10ml Posiflush™ syringes, using a rapid, pulsating push / pause positive pressure technique until line is clear to prevent the risk of occlusion and infection.[4,5,10,16-18]
d. Attach Pink Vacutainer® transfer device to syringe. Insert appropriate blood tubes in correct order of draw as required

Alternate sampling method using direct connection of Vacutainer® to PPV

a. Attach Blue Vacutainer® access device to PPV.
b. Insert appropriate blood tubes in correct order of draw as required.
c. Immediately following blood sampling, scrub the PPV with a large 70% Alcohol impregnated wipe and flush with 20ml sodium chloride 0.9% using 2x10ml Posiflush™ syringes, using a rapid, pulsating push / pause positive pressure technique until line is clear to prevent risk of occlusion and infection.[4, 5,10,16-18]

- Dispose of sharps and waste appropriately.
- Remove gloves and perform hand hygiene.
- If used, clean trolley with combined disinfectant detergent wipes.

- Perform hand hygiene.
- Label and sign samples, complete request form and send to Pathology.

8. Removal

8.1 Medical Requirements

8.1.1 Daily medical review of need for continued central vascular access. Remove CVC or PICC promptly when no longer clinically indicated.[3]

8.1.2 CVC or PICC are not to be routinely replaced. There is no high-level evidence to support this practice.[4,5]

8.1.3 In cases of suspected infection, contact Infectious Diseases and the SCGH IV Access Service. Obtain full septic screen including:

- Two sets of blood cultures – one set of blood cultures drawn through the CVC or PICC, and a second set of blood cultures drawn from a peripheral vein at the same time
- Wound swab from exit site.
- If catheter is to be removed, send tip for culture.

8.2 Safety Information

8.2.1 Refer to General Safety Information section: Page 3

8.2.2 Ideally, the platelet count should be more than 50 × 10^9/l and the INR less than 1.5 before catheter removal.[22]

8.2.3 To reduce the risk of air emboli:

- For CVC removal – Lie the patient flat.[23]
- For PICC removal – Position patient supine with arm extended 45-90 degrees. The PICC insertion site should be below the level of the heart.[23,24]

8.2.4 Routine removal of a CVC or PICC device should only occur during business hours when resources are easily accessible.

8.2.5 If CVC or PICC removal is due to suspected infection, ensure catheter tip is sent for culture.

8.2.6 Carefully inspect catheter after removal to ensure it is complete.[22,25]

8.3 Procedural Information

8.3.1 **Equipment Requirements**

- PPE including non-sterile and sterile gloves
- Dressing pack
- Chlorhexidine 2% in Alcohol 70% or Prontosan solution
- Sterile occlusive dressing
- Stitch cutter (for CVC removal)
- Adhesive remover wipe
- Sterile scissors and specimen container if sending catheter tip to pathology

8.3.2 **Procedure**

- Confirm patient's identity and explain the procedure[23]
- Perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes. Allow to dry
- Gather all equipment.
- Perform hand hygiene and prepare the sterile field.
- Open dressing pack using the corners and add equipment to the sterile field using aseptic non touch technique.
- Perform hand hygiene.
- Position the patient.
- Perform hand hygiene and put on non-sterile gloves.
- Remove CVC or PICC dressing using non touch technique using adhesive remover wipe if required.
- Remove gloves.

- Perform hand hygiene and put on sterile gloves.
- Protect key parts using a non-touch technique. Clean around insertion site with Chlorhexidine 2% in Alcohol 70% or Prontosan solution using aseptic non touch technique. Allow to air dry for 30 seconds..[8]
- CVC removal:
  a) Remove sutures.
  b) Instruct patient to take 2 deep breaths and hold the third breath as catheter is removed. This increases thoracic pressure and reduces the risk of an air embolism.[22,23]
- PICC removal:
  a) Loosen securing device.
  b) Ensure the arm is supported and abducted 45-90 degrees. Grasp PICC at insertion site and slowly remove using a 2-5cm stroke-pause-stroke action. Avoid pressure to the insertion site.[24]
- Encountering resistance:
  a) Generally PICCs can be removed with little resistance
  b) If resistance occurs, pause for 10-30 seconds before trying again.
  c) If resistance continues, consider relaxation techniques such as; apply a warm compress to the insertion site and upper arm, offer a warm beverage, deep breathing exercise, administer a slow sodium chloride 0.9% flush then try again.[23]
  d) If resistance persists contact the SCGH IV Access Service or After Hours CNS.
- Following removal of CVC or PICC:
  a) Immediately apply firm pressure using sterile gauze over exit site until bleeding ceases.[23-25]
  b) Leave pressure gauze in situ and apply an occlusive dressing over exit wound. Leave intact for 24 hours.[23,25]
- In cases of suspected infection:
  a) Obtain wound swab of exudate from exit site. Send to Pathology for culture and sensitivity testing.
  b) Cut approx. 4cm from the CVC or PICC catheter tip using sterile scissors. Place in a sterile specimen container and send to Pathology for culture testing.
  c) Carefully inspect catheter to ensure it is intact[22,25] (refer to insertion details if unsure of catheter type or contact SCGH IV Access Service).
- Dispose of sharps and waste appropriately.
- Remove PPE including sterile gloves and perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes.
- Perform hand hygiene.
- Document removal on **SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR864.2)** or **OPH Intravascular Device Insertion Record (MR 122)** and medical record. If at **SCGH** also fax copy to the SCGH IV Access Service (Fax No: 6457 4375).[23]

9. Unblocking Lumens with Alteplase

9.1.1 It is a legal requirement that all disciplines that prescribe or administer medications comply with the Medicines and Poisons Act 2014, Medicines and Poisons Regulations 2016, Pharmacy Act 2010 and Hospital Guidelines. It is a legislative requirement that a valid prescription must exist before Schedule 4 and Schedule 8 medications are administered.

9.1.2 Only valid prescriptions will be utilised by any nursing staff to administer medication except for verbal orders and nurse-initiated medications, as per designated list.

9.1.3 A valid prescription must include:

- Doctor’s or Nurse Practitioner’s signature PLUS printed surname or stamp of surname
- A prescriber number is necessary to prescribe medications (Schedule 4 and above).
- The name and address of the patient
- The date when it was written
- The name, form, strength, and quantity of the medication, and
- Directions for use, including route, dose and frequency

9.1.4 A nurse who is registered as a nurse practitioner with the Nurses and Midwives Board of WA and is working in a designated area may prescribe medications from approved clinical protocols.

9.2 Medical Requirements

9.2.1 Alteplase must be prescribed on the medication chart by an authorised prescriber as a once-off medication, specifying the dose, volume, device and intended lumen.

9.3 Safety Information

9.3.1 Refer to General Safety Information section: Page 3. Appropriate PPE must be used to prevent contamination and mucosal or conjunctival splash injuries.

9.3.2 Alteplase is available as a 2mg/2ml syringe and is indicated for occluded catheters to promote patency of the central venous access device and to resolve clotting.[12-15] **ONLY USE THE 2mg/2ml SYRINGES FOR OCCLUDED CATHETERS**.

9.3.3 **Do not use** Alteplase

- If a catheter-associated infection is suspected.
- To unblock lumens being used for infusions containing inotropic or vasoactive drugs. Lines with inotropic or vasoactive drugs are not routinely flushed hence use of Alteplase may unintentionally expose patients to inotropic or vasoactive drugs left in line.
- If unable to aspirate / inject any fluid into the line there is no indication for Alteplase. Contact the SCGH IV Access Service for advice.

9.3.4 Potential complications of use include:

- Increased anticoagulant effect
- Haemorrhage
  Note: any Alteplase released into the circulation is metabolised rapidly by the liver (plasma half-life = 5 mins) so risk of bleeding is minimal.
- Anaphylaxis (rare < 2%)
- Rapid lysis of coronary artery thrombi causing cardiac arrhythmias (MIMS online, 2008)

9.4 Management

9.4.1 Exclude other possible causes of occlusion i.e. kinked catheter, incorrect position or clamped tubing.

9.4.2 Try clearing the line with a 10mL Sodium Chloride 0.9% flush, using a vigorous, pulsating (stop-start) technique. If unable to inject any fluid into the line, there is no indication for Alteplase. Contact the SCGH IV Access Service for advice.

9.4.3 Alteplase is stored in the freezer and needs to be thawed for 1 hour prior to administration. Once removed from the freezer, the syringe must be used within 24 hours.

9.4.4 Alteplase must be checked by 2 nurses, one of whom must be a Registered Nurse. Administration is to be conducted by a Registered Nurse only.

OPH specific information*: Alteplase 2mg/2ml syringes are stored in the Ward 1 freezer. Please contact Pharmacy in the first instance to arrange supply unless it is required after-hours.*

9.4.5 Dosage is determined by the lumen priming volume of the catheter. For CVC and PICC the lumen priming volume is 1mL.

- Unblocking one <u>lumen</u> of a single or multiple lumen CVC or PICC:
  Administer a volume of 1mL of reconstituted Alteplase into the affected lumen.
- Unblocking <u>two or more lumens</u> of multiple lumen CVC or PICC:
  Administer a volume of 1mL of reconstituted Alteplase into each affected lumen.
- If all lumens are sluggish:
  Administer a volume of 1mL of reconstituted Alteplase into the distal lumen.

9.4.6 Repeat administration of Alteplase within a 24 hour period if lumen remains sluggish.

9.5 Procedural Information

PART 1: PREPARATION AND INSTILLATION OF ALTEPLASE

9.5.1 Equipment Requirements

- Disposable kidney dish
- PPE including non-sterile gloves
- Large 70% Alcohol impregnated wipe
- 2 x Sodium Chloride 0.9% ampoules and 2 x 10mL luer lock syringe and drawing up needle or 2 x BD Posiflush™ Pre-filled Saline Syringes[10,17,18]
- 1 x Alteplase 2mg/2ml syringe
- 1 x Water for Injection ampoule

9.5.2 Procedure

- Remove the Alteplase from the freezer at least 1 hour before administration to allow it to defrost.
- Perform hand hygiene.[6]
- Clean dressing trolley with combined disinfectant detergent wipes.
- Gather equipment for the procedure.
- Perform hand hygiene.[6]
- Perform hand hygiene and apply non-sterile gloves.
- Scrub the lumen using a non-touch technique with a large 70% Alcohol impregnated wipe for 30 seconds using different areas of the wipe.[6] Allow to air dry for 30 seconds.
- Flush with Sodium Chloride 0.9% for Injection using a 10mL syringe, or BD Posiflush™ Pre-filled Saline syringe.
- Using non-touch technique, instil Alteplase 1mL slowly using a gentle pulsating action. **Note**: this may take some time. Instil a small amount at a time.
- Label the device with a sticker indicating Alteplase insitu.[7]
- Dispose of waste appropriately.
- Remove gloves and perform hand hygiene.
- Clean trolley with combined disinfectant detergent wipes.
- Perform hand hygiene.
- Document date and time of instillation.
- Allow to dwell for a minimum of 1 hour before attempting to aspirate. Ideally leave for 4 hours or, optimally, up to 24 hours.

PART 2: ASPIRATION OF ALTEPLASE

9.5.3 Equipment Requirements

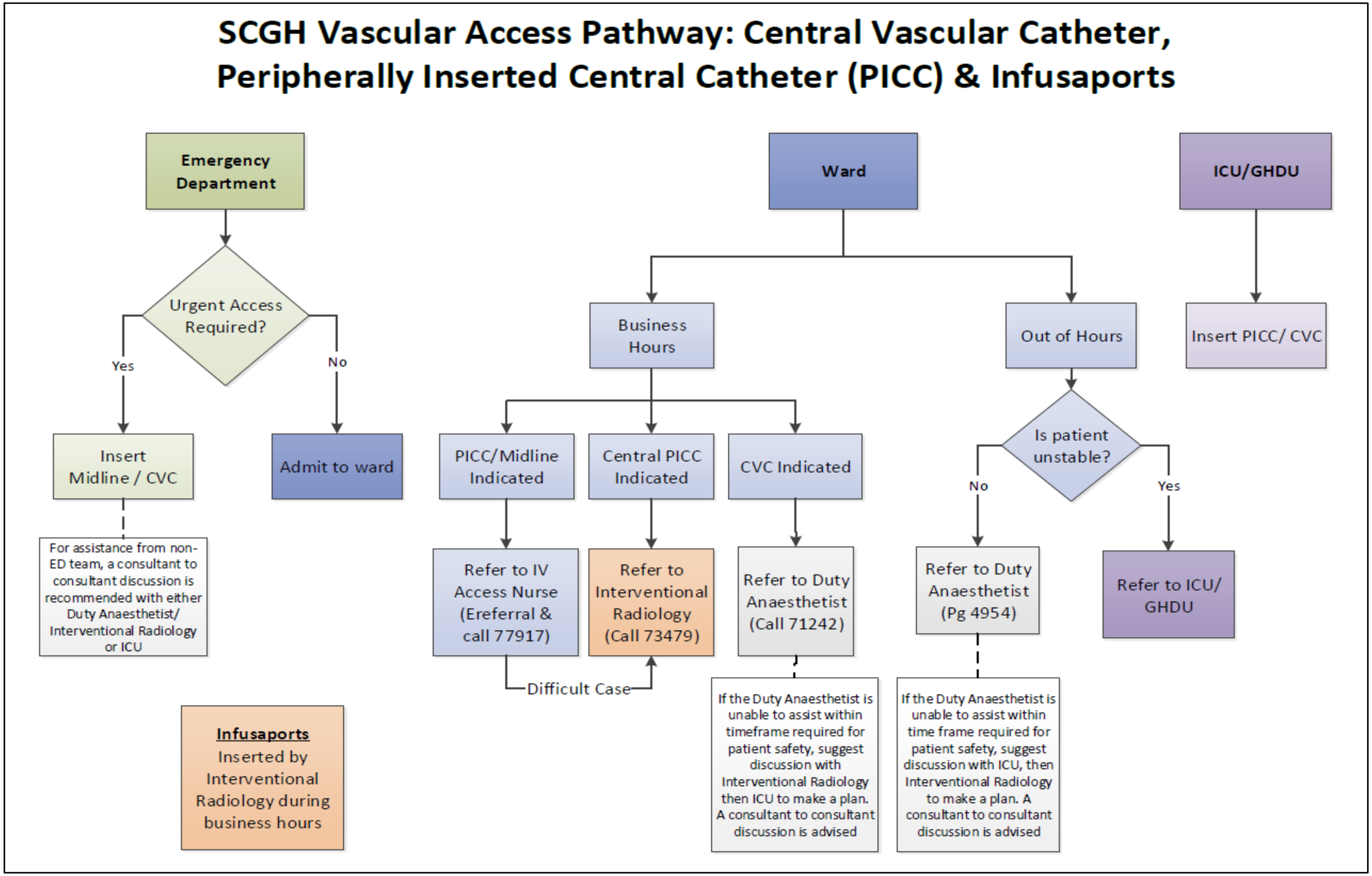
- Disposable kidney dish
- PPE including non-sterile gloves
- Large 70% Alcohol impregnated wipe
- 2 x Sodium Chloride 0.9% ampoules and 2 x 10mL luer lock syringes and drawing up needles or 2 x BD Posiflush™ Pre-filled Saline Syringes[10,17,18]
- 1 x 5mL Luer lock syringe

9.5.4 Procedure

- Perform hand hygiene.[6]
- Clean dressing trolley with combined disinfectant detergent wipes.
- Gather equipment for the procedure.
- Perform hand hygiene[6] and apply non-sterile gloves and other PPE as indicated.
- Scrub the lumen using a non-touch technique with a large 70% Alcohol impregnated wipe for 30 seconds using different areas of the wipe.[6] Allow to air dry for 30 seconds.
- Attempt to aspirate Alteplase from lumen with 5mL syringe. If unable to aspirate, the Alteplase may be safety flushed into the catheter.
- Once the blockage is cleared, flush well with 10-20mLs Sodium Chloride 0.9%.

- Dispose of waste and remove gloves.
- Perform hand hygiene.[1,7,8,11]
- Clean trolley with combined disinfectant detergent wipes
- Perform hand hygiene
- Document in patient's medical record.
- If the catheter remains occluded, contact the SCGH IV Access Service for advice.

Appendix 1: Vascular Pathways for Insertion of CVC / PICC



Appendix 2: CVC Troubleshooting Guide

Complications	Signs and symptoms	During insertion	Post insertion
Air embolism / catheter disconnect	• Restlessness or anxiety Dyspnoea /Cyanosis • Hypertension/dizziness • Tachycardia/week pulse • Altered consciousness (e.g. confusion) • Visible catheter damage or leakage from line during use	During insertion: • Position patient supine (with head down 30-degree tilt where possible) • Ensure all CVC connections are secure	• Check all CVC connections and prevent further air entry (cap line with new positive pressure valve) • Position patient in left Trendelenburg (head down at 30-degree tilt) • Administer high flow oxygen • Seek urgent medical officer (MO) review • Monitor vital signs • Complete Clinical Incident Form **If visible damage:** • Clamp CVC between patient and damaged area, cover with sterile gauze • Minimise patient movement • Seek medical review and expert advice on management **NB**. Removal or repair of tunnelled catheters should only be undertaken by specialist staff
Arterial puncture	• Bright red blood • Syringe fills more quickly than expected (when compared to venous) • Blood leaves in a pulsing mode	• Apply direct pressure to site for 5 minutes to limit haematoma formation • Monitor vital signs • Document in medical record • Inform medical staff if tachycardia or hypotension occurs	N/A
Cardiac tamponade	• Chest tightness or pain • Shortness of breath • Muffled heart sounds • Altered consciousness (as tamponade impairs myocardial contractility and result in loss of cardiac output).	• Administer high flow oxygen • Monitor vital signs • Documented in medical record • Activate MET • Prepare for peri-cardiocentesis (as directed by MO)	• Monitor clinical conditions and vital signs as indicated • Request MO review as indicated by patient's condition
Catheter dislodgement	• External portion of catheter is visibly longer • Sudden lack of blood return • Swelling in arm, neck or chest	N/A	• Do not attempt to push catheter back into place or use it • Seek guidance from MO

Complication	Signs and symptoms	During insertion	Post insertion

Catheter occlusion	• Inability to flush catheter • Blood in lumen • Inability to aspirate blood Note: partial obstruction may allow infusion of fluid but not aspiration **Cause of internal occlusion:** • CVC migration to smaller vein or resting on vein wall • Compression of CVC in superior vena cava between the clavicle and the first rib • Constant wearing may result in damage to the integrity of the catheter • Other causes include chemical precipitation, lipid disposition or thrombus formation		• Routine use of push/pause technique for lumen flushes **External occlusion:** Check for kinks (or clamp) and reassure line if indicated **Internal occlusion** • Ask patient to cough, take deep breaths, hunch shoulders or reposition • Attempt gentle irrigation with sodium chloride 0.9% using push/pause method alternating flushing and aspirating; do not exert excessive pressure or suction • If unable to remedy, or patient has signs or symptoms of a thrombus seek urgent MO review • CXR may be required to confirm tip position/exclude migration • If thrombus suspected – see below
Haematoma formation	• Local swelling • Patient complains of pressure	• Record and monitor symptoms • Seek medical help if condition advances	• Same as for during insertion • Vigilance is required in patients with coagulopathy
Infection	**Site infection and tunnel infection:** • Local pain or inflammation • Cellulitis or tracking within 2cm of exit site • Exudate or pus at site • Fever **Systemic infection:** • Fever (usually >38°C) • Malaise • Rigors (particularly on flushing line) • Chills • Hypotension • Other signs of shock may develop so close observation is required	Compliance with CVC Bundles: • Maximal barrier precautions on insertion • Hand hygiene • Skin anti-sepsis with Chlorhexidine 2% in Alcohol 70% • Optimal selection insertion site; avoid femoral approach • Use Chlorhexidine-impregnated dressing (Tegaderm CHG™ or BIOPATCH®)to reduce the regrowth of microbes after skin antisepsis	• Review need for CVC daily • Other routine care including hand hygiene, site monitoring, line change/lumen flushing as detailed in this clinical practice guideline **If local infection suspected:** • Inform MO and Shift Coordinator, review management plan • Review need for CVC • Take swab of site if exudate present • Consider antibiotic therapy • Document in medical record **If signs of systemic infection:** • Inform MO and Shift Coordinator and review management plan • Monitor closely for signs of shock • Remove CVC if no longer needed, and send tip for culture • Obtain two sets of blood cultures (from CVC and peripheral vein) prior to commencing antibiotics.

Complication	Signs and symptoms	During insertion	Post insertion
Pneumothorax	• Restlessness or anxiety • Dyspnoea • Cyanosis • Pain on breathing • Oxygen desaturation • Wheeze	• Monitor patient's respiratory rate and general condition • If concerned, discontinue procedure • Obtain urgent CXR	• Monitor patient's respiratory rate and general condition • Routine post insertion CXR if subclavian or jugular vein insertion **If signs or symptoms evident:** • Sit patient upright • Administer high flow oxygen • Seek urgent medical review
Thrombus	• Swelling • Pain, numbness or tingling • Coolness, swelling or venous engorgement, discoloration of neck, chest or arm (on side of CVC)		• Patients at high risk of CVC thrombus (e.g. patients with cancer) maybe prescribed routine anticoagulation • Seek urgent MO review • Monitor vital signs • Consider need for oxygen therapy (if appropriate) • Consider fibrinolytic agent (e.g. Alteplase) as prescribed by MO

Source: South Metropolitan Health Service (SMHS) 2014 SMHA Vascular Access Device Management Clinical Practice Standard. Reproduced with permission. Refer to original document for supporting evidence.

Appendix 3: PICC Troubleshooting Guide

Signs/Symptoms	Possible Causes	Suggested Actions
Bleeding at exit site	• Difficult insertion/ number of punctures at site • Coagulation abnormalities • Consider comorbidities, e.g. patients with liver abnormalities may continue to bleed following insertion	• Apply digital pressure or pressure dressing to site for 5-15 minutes or as clinically indicated • Consider topical thrombin dressing as prescribed • Correction of coagulation – one or all the following may be given: platelets/FFP/Vitamin K as prescribed • Document bleeding in health record • Inform MO or SCGH IV Access Service and shift coordinator if prolonged bleeding
Catheter Occlusion Unable to flush lumen	**External** • Kinked or clamped catheter **Internal** • Chemical precipitation - can occur gradually causing sluggish flow or immediately causing total occlusion • Lipid deposition • Crystallisation of Total Parental Nutrition (TPN) mixture or drug incompatibilities	• Ensure line is unclamped • Ensure positive pressure valve in situ • Take down dressing and check for kinks. Realign if necessary • Resecure lumen/line and apply new dressing • Check for drug compatibilities • Use dedicated lumen for TPN • Ensure regular flushing of lumens, for inpatients • Contact MO or SCGH IV Access Service
Catheter Occlusion May observe blood or precipitate in lumen	• Thrombus formation can originate outside or inside the catheter lumen • Causes include venous stasis, vessel wall trauma or stenosis hypercoagulable states and performing blood pressure readings (utilizing a cuff) on the arm with a PICC in situ. • Cuff pressure can cause bleeding at the insertion site, increasing the risk of thrombus formation and cause retrograde blood flow raising the risk of catheter occlusion.	• Ensure regular flushing of lumens, using 10mL sodium chloride 0.9% and push/pause technique until blood or precipitate is removed. • Ensure positive pressure valve is correct for device • Contact MO or SCGH IV Access Service • If unsuccessful, consider fibrinolytic agent (e.g. Alteplase) as ordered by MO.

Signs/Symptoms	Possible Causes	Suggested Actions
Infection • Pyrexia temperature • 38 degrees +/- rigors after flushing lumens • Tachycardia • Hypotension • Shock	• Catheter Related Blood Stream Infection	• Liaise with MO or SCGH IV Access Service - consider removal of PICC and/or alternative IV access and send tip for culture • Document findings and actions taken in patient medical records • Take 2 sets of blood cultures (from PICC line and peripheral vein). • Monitor patient TPR, BP, Sp02, responsiveness – frequency determined by patient condition • Activate MET response.
Insertion Site Inflammation, redness, tenderness +/- discharge at exit site	• Infection	• Take swab of exit site • Notify MO and document in patient integrated notes • Consider removal of PICC – Liaise with MO or SCGH IV Access Service • Ensure insertion site is dressed with Chlorhexidine-impregnated dressing (Tegaderm CHG™ or BIOPATCH®) • Contact SCGH IV Access Service to review insertion site
Pain, visible swelling or fluid leaking from site: when PICC is flushed	• Malposition of catheter • Fibrin sheath • Internal or external catheter fracture • Extravasation • Thrombus	• Stop infusion of fluids • Notify MO and document in patients integrated notes • Chest x-ray or ultrasound may be ordered • Consider removal of PICC – Liaise with MO • If confirmed thrombus, follow Central Catheter Related Venous Thrombosis (CCRVT) Treatment Guideline Recommendations. Notify Anticoagulation CNC.
Partial Occlusion • Reduced flow of infusion • Difficulty in flushing lumen(s)	• Tip resting against a vein valve/wall • Fibrin sheath formation • Position of catheter	• Request patient to cough, take deep breaths, hunch shoulders, or change position. • Document in patient integrated notes • Liaise with MO or SCGH IV Access Service • PICC may be rewired within 24 hours by SCGH IV Access Service to avoid reinsertion if malfunction present without infection.
PICC line migration	• Over exertion, e.g. excessive coughing • Lifting heavy weights • Dressing insecure, or during dressing changes • Unknown	• Access line and compare to insertion details • Contact MO or SCGH IV Access Service

Source: South Metropolitan Health Service (SMHS) 2014 SMHA Vascular Access Device Management Clinical Practice Standard. Reproduced with permission. Refer to original document for supporting evidence.

Appendix 4 – Consent Script

Guideline for obtaining informed consent for a Peripherally Inserted Central Catheter (PICC)

Peripherally Inserted Central Catheter (PICC)	
Scope	Nursing staff of the IV Access Service or Medical Practitioner.
Governance	Nursing staff of the IV Access Service can obtain informed consent from patients (or their guardian) for insertion of a Peripherally Inserted Central Catheter (PICC). Endorsed by the Clinical Governance Committee [04/12/2024]. Consent to Treatment (Form A) MR 730.81 must be used. Details of the consenting discussion should also be recorded in the patient's integrated health record.
Purpose of this document	To ensure a standardised and consistent approach to the consenting discussion as led by the IV Access Nurse obtaining consent. This document should be read in conjunction with the following: **https://healthpoint.hdwa.health.wa.gov.au/policies/Policies/NMAHS/SCGH/SCGOPHCG.HP.Consent.pdf**
PICC Device	A PICC is a Peripherally Inserted Central Catheter. It is a long and narrow catheter that is inserted into a vein in the upper arm, above the elbow, using ultrasound guidance. A PICC can be used to administer intravenous medication, blood and other blood products, intravenous fluids, chemotherapy, and Total Parenteral Nutrition (TPN). A PICC can also be used for blood sampling.
PICC Insertion Procedure	• The PICC is inserted by a specially trained nurse within the IV Access Service, in the ground floor Radiology department or at the bedside, as required. • The procedure to insert a PICC takes approximately one hour from the time the patient arrives in the procedure area. • During the PICC insertion the patient will be required to lie on their back with their arm extended out to the side, preferably at 90 degrees. • The patient will be awake for the procedure; however local anaesthetic will be used to numb the area and minimise any discomfort. • Once a suitable vein has been identified, the skin around the area will be cleaned and the local anaesthetic injected. • The catheter is inserted in the upper arm and carefully advanced within the vein and towards the heart. • A sterile dressing is placed over the insertion site and a cap placed over the end of the catheter to keep it free from bacteria at the conclusion of the procedure.

<table>
<tr><td></td><td>When it's thought the catheter tip has reached the ideal location (anywhere between the lower third of the Superior Vena Cava and the top third of the Atrium) a chest x-ray will be performed to verify correct catheter placement.</td></tr>
<tr><td>Benefits and Risks of a PICC</td><td>

Confirm with the patient that they are aware they have been referred by their doctor for a PICC.
Confirm with the patient that they are aware of the reason for a PICC.
Inform the patient of the benefits and risks as outlined below.

Benefits:

A PICC can be used to deliver medications and other treatments directly into the blood stream and into the larger veins near the heart.
A PICC can be used for blood sampling omitting the need for needles for blood testing.
A PICC can be left in place for many weeks or months, depending on the treatment needs, versus a Peripheral Intravenous Catheter (PIVC) which must be changed every 72 hours.
A PICC can allow for a discharge from hospital when clinically indicated and the completion of remaining therapy within the community.

Risks and/or Complications: As with all invasive procedures, some risks are inherent in the placement of a PICC.

Pain, bruising, and/or bleeding can occur at the insertion site.
Difficulty inserting the PICC.
Infection at the PICC insertion site which may lead to an infection within the bloodstream.
Inflammation of the vein (Phlebitis).
Development of a blood clot (Thrombosis).
Temporary nerve pain and/or damage.
Dysrhythmias of the heart if the catheter or wire enter the heart.
Arterial puncture.
Damage to blood vessels.
Air embolism
Pneumothorax or collapsed lung. Very RARE.

Some complications relating to a PICC insertion can be managed so that the PICC can remain in place. (i.e. Anticoagulation for confirmed thrombus, as prescribed).
Some complications may require early removal of a PICC (i.e. confirmed infection).
</td></tr>
</table>

Appendix 5 – Vein Assessment Tool

Peripherally Inserted Central Catheter (PICC) Vein Assessment Tool		
Patient suitable for a peripheral PICC	YES | NO | Single | Double	
Comment:		
Suitable Vessel/s:		
(R) Basilic	Diameter:	mm
(R) Brachial	Diameter:	mm
(R) Cephalic	Diameter:	mm
(L) Basilic	Diameter:	mm
(L) Brachial	Diameter:	mm
(L) Cephalic	Diameter:	mm
Consent completed	YES | NO	

Compliance, monitoring and evaluation

All clinical staff who perform aseptic procedures have a responsibility to perform Aseptic Technique according to hospital policies, procedures and protocols.

HODs in each discipline are responsible for ensuring all clinical staff who perform aseptic procedures are trained and assessed competent in Aseptic Technique. Compliance with Aseptic Technique e-Learning and Competency Assessment is regularly monitored by HODs.

Compliance with this multidisciplinary Clinical Practice Guideline will be monitored through routine clinical incident review process.

Related Documents

- ANZICS 2012, Central Line Insertion and Maintenance Guideline
- WA Health Consent to Treatment Policy
- NMHS Consent to Treatment Policy
- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- Pharmacy Act 2010
- Work Health and Safety Act 2020
- NMHS Clinical Documentation Policy
- National Standard for User-applied Labelling of Injectable Medicines Fluids and Line
- OPH Infection Control Practice Guidelines – A5 Aseptic Technique
- SCGH Infection Prevention and Control Manual - Aseptic Technique Policy #18
- SCGH Nursing Practice Guideline #4: Intravenous Therapy
- Central Catheter Related Venous Thrombosis (CCRVT) Treatment Guideline Recommendations
- SCGOPHCG Clinical Practice Guideline Vein Selection and Preservation in Patients with Chronic Kidney Disease
- Work-Health-and-Safety-Management-Policy

Other supporting documents

- SCGH Intravascular Device Insertion Record Central Vascular Access Device (MR864.2)
- OPH Central Vascular Access Device Insertion record (MR122)
- SCGH Infection Prevention and Control Guideline: Reprocessing Ultrasound Transducers

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Applicable Standards	NSQHS Standards 1: Clinical Governance Std 3: Preventing and Controlling Healthcare Associated Infection Std 4: Medication Safety Std 5: Comprehensive Care Std 6: Communicating for Safety Std 7: Blood Management Std 8: Recognising and Responding to Acute Deterioration				
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